



Komputer Teknik dan Internet of Things

Predictive Analytics

Python Dashboard

Hamsar

Smart Industries

Asset Management

📁 Data Source

Select input method:

- ☒ 📁 Upload File
- ☐ 🔗 Enter File Path

Drop or select Excel file

📤 Upload

200MB per file • XLSX, XLS



Dashboard Pemakaian Bahan Kimia - Water Treatment Plant ➡

📅 Data diperbarui otomatis setiap pukul 06:00 WIB | 🔍 Analisa tren berbasis AI-Heuristic

📅 Rentang Waktu

2026/04/05 – 2026/05/04

🔧 Bahan Kimia

PAC ✕

Polimer Anionik ✕

Kapur (Lime) ✕

Klorin ✕



Ringkasan Kinerja

Total Pemakaian

10,145 kg/L

📈 338/hari

Rata-rata Kekeruhan Baku

6.1 NTU

Volume Terolah

4,725 m³/hari

Estimasi Biaya

Rp 152,177,341



Tren Pemakaian Harian

Data Source

Select input method:

Upload File

Enter File Path

Drop or select Excel file

data_daily_wtp.xlsx

11.5KB

+

Date Configuration

Choose Date Column:

Tanggal

Select date range (leave as default for full range):

2026/02/09 – 2026/05/04

The chart displays four stacked data series: PAC (dark blue), Polimer Anionik (light blue), Kapur (red), and Klorin (pink). The total value fluctuates between approximately 150 and 380 units. PAC consistently forms the largest base of each bar, followed by Polimer Anionik, Kapur, and Klorin at the top. The x-axis labels include Apr 5 2026, Apr 12, Apr 19, Apr 26, and May 3.

Interactive Excel Dashboard

Loaded as rows from data_daily_wtp.xlsx

Showing 85 of 85 rows (2026-02-09 → 2026-05-04)

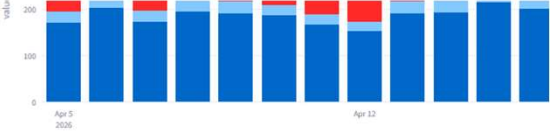
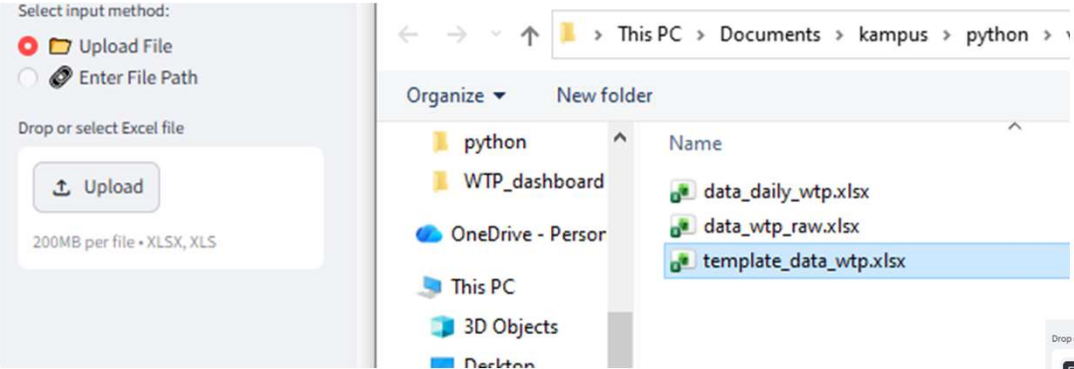
Data Preview

Your Charts/Logic

Replace this section with your existing dashboard code. Use `df_filtered` as your data source.

Analisa & Rekomendasi Berbasis AI

- PAC: Stabil (perubahan -5.8%).
- Polimer Anionik: Stabil (perubahan -5.7%).
- Kapur (Lime): Stabil (perubahan -0.9%).
- Klorin: Stabil (perubahan +6.6%).
- Korelasi PAC vs Kekeruhan: positif kuat ($r=0.73$). Sistem dosing manual rentan over/under-dose. Rekomendasi: pasang turbidimeter online + PLC feedback control.
- Dosis PAC 7 hari terakhir: 38.97 mg/L. Perlu penyesuaian.
- Tren PAC menurun. Evaluasi apakah terjadi bypass, kebocoran pipa, atau perubahan karakteristik air baku.



Interactive Excel Dashboard

✓ Loaded 43 rows from template_data_wtp.xlsx

Showing 43 of 43 rows (2026-03-23 → 2026-05-04)

Data Preview Your Charts/Logic

	Tanggal	PAC	Polimer Anionik
0	2026-03-23 00:00:00		177
1	2026-03-24 00:00:00		225
2	2026-03-25 00:00:00		199
3	2026-03-26 00:00:00		135
4	2026-03-27 00:00:00		213
5	2026-03-28 00:00:00		178
6	2026-03-29 00:00:00		224
7	2026-03-30 00:00:00		210
8	2026-03-31 00:00:00		145
9	2026-04-01 00:00:00		173

Download Filtered CSV

WTP_dashboard > dashboard_wtp.py > ...

```
1 import streamlit as st
2 import pandas as pd
3 import plotly.express as px
4 import plotly.graph_objects as go
5 from datetime import datetime, timedelta
6 import numpy as np
7 import os
8
9 # ===== CONFIG =====
10 DATA_FILE = "wtp_chemical_usage.csv"
11 CHEMICALS = ["PAC", "Polimer Anionik", "Kapur (Lime)", "Klorin"]
12 UNIT_COST_PER_KG = 15_000 # IDR (contoh)
13
14 # ===== DATA ENGINE =====
15 def generate_mock_data(days=30):
16     np.random.seed(42)
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, **/*.ts, !**/nod



AI Kraft Pulp Cooking Dashboard

Batch & Continuous Digester Optimization | Kappa & H-Factor Control

Batch Cooking

Continuous Cooking

Upload Process Data (.xlsx)

Choose File cooking_data.xlsx

Process & Load Dashboard

Loaded 6 batch runs successfully.

Avg Kappa Number

29.15

Avg H-Factor

1491.3

Steam Efficiency

937.9%

Optimal Batches

0 / 6

Kappa Number Trend & Delignification Control



Steam Demand & Energy Efficiency



📄 Batch/Run Parameters

⚠️ Process Alerts & AI Recommendations

- ◆ Batch 1: High steam_pressure: 42.0
- ◆ Batch 1: High chip_moisture: 485.0
- ◆ Batch 2: High steam_pressure: 45.0
- ◆ Batch 2: High chip_moisture: 510.0
- ◆ Batch 3: High steam_pressure: 43.0
- ◆ Batch 3: High chip_moisture: 475.0
- ◆ Batch 4: High steam_pressure: 45.0
- ◆ Batch 4: High chip_moisture: 529.0
- ◆ Batch 5: High steam_pressure: 41.0
- ◆ Batch 5: High chip_moisture: 469.0
- ◆ Batch 6: High steam_pressure: 45.0
- ◆ Batch 6: High chip_moisture: 500.0

#	Chip BD (kg/m ³)	Moisture (%)	WL Strength (g/L)	BL AA (g/L)	WL Temp (°C)	Cook Temp (°C)	Steam P (bar)	Steam T (°C)	H-Factor	Pred. Kappa	Steam Eff.
1	450	485	115	223	95	165	42	152	1450	30.45	911.84
2	462	510	118	231	97	168	45	155	1520	30.41	975.48
3	453	475	115	243	96	166	43	153	1459	26.54	933.07
4	442	529	116	237	97	167	45	155	1535	30.85	969.68
5	457	469	115	228	98	165	41	157	1455	28.36	861.78
6	454	500	117	239	97	168	45	155	1529	28.3	975.48

renv > cooking > app.py > ...

```
1 import os
2 import numpy as np
3 import pandas as pd
4 from flask import Flask, render_template, request, jsonify
5 from datetime import datetime
6
7 app = Flask(__name__)
8 app.config['UPLOAD_FOLDER'] = 'uploads'
9 os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)
10
11 # Required columns from your prompt
12 REQUIRED_COLS = [
13     "chip_bulk_density", "chip_moisture", "white_liquor_strength",
14     "black_liquor_active_alkali", "white_liquor_temperature",
15     "cooking_temperature", "steam_pressure", "steam_temperature", "h_factor"
16 ]
```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Filter (e.g. text, **/*.ts, !*/nod...)



